

purpose. Last year, the US Air Force took into consideration the risks taken by soldiers during re-supply missions and started using an unmanned helicopter — K-MAX built by Lockheed Martin — in places like Afghanistan.

As far as ground robots are concerned, there is still a lot of work going on in this area. But for the purpose of defence, the robots are mainly used in a tele-operational capacity. This is because tele-operated robots are more easily controlled in missions like bomb diffusion.

"There certainly may be a time when unmanned vehicles rule the day, but not for a few decades to come as reliability is a prime concern and such technologies are still in their nascent stages of development," says Pitchai.

T. Jayakrishnan, an experienced roboticist and consultant to US-based robotics companies, also agrees with the benefits of unmanned robotic systems. "Unmanned robots are particularly helpful for detecting hazards such as roadside bombs and chemical-biological threats. Using robots means that people can remain at a safe distance while they check a road or building for hazards," he says.

Some of the unmanned robots that Jayakrishnan admires are Armadillo (a micro-UGV that can be thrown) and LandShark (a mid-size UGV). The latter features an open platform for multiple payloads (sensors, arm, digger, etc), tows 200 kg or lifts 70 kg with a heavy-lift arm, and can carry and drop-off/retrieve multiple micro-UGVs. Jayakrishnan also highlights to notice the Oshkosh M-ATV, which is a manned or unmanned larger vehicle for rescue and cargo.

Massachusetts-based iRobot Corp revealed pre-production prototypes of another small and light ground robot for military missions at the *Consumer Electronics Show* last year. The 110 FirstLook UGV provides hasty situational awareness, performs persistent observation and investigates confined

Defence robotics at a glance

Push factors:

- > Strong desire to reduce or prevent military casualties in the field of operations
- > Changes in the tactics of warfare requiring new reconnaissance
- > Combat, task machinery and tools
- > The need to reduce military spending
- > Developments in the fields of materials science, computer programming and sensing technology to help create more advanced robots

Pull factors:

- > Weak economic conditions
- > Dearth of active military conflicts for most of the world, which reduces the need for new defence systems
- > Ethical concerns involving the use of robots for war-fighting operations

—Adapted from an ABI Research report titled "Defense Robots: UAVs, UGVs, UUVs and Task Robots for Military Applications"

spaces. It weighs less than 3 kg and is 25.4 cm (10 inches) long, making it ideal for building clearing, raids and other close-in scenarios.

The company believes that FirstLook will be an important tool for war-fighters and first responders. The machine is equipped with four built-in cameras and two-way audio communication. It also has digital mesh networking capabilities, allowing multiple robots to relay radio communications over greater distances. The device uses

a wrist-mounted, touch-screen operator control unit with an inbuilt radio.

iRobot, in fact, has a large range of products being used in war, including the 510 PackBot, small unmanned ground vehicle (SUGV), 710 Warrior and the Negotiator.

"SUGV is a tactile mobile robot that gathers situational awareness in dangerous conditions for war-fighters and public safety professionals. It is lightweight and can operate in a variety of conditions and weathers, as well as on rough terrain and in urban settings, hence commonly used for surveillance or reconnaissance, bomb disposal, explosive detection and inspections," explains N.H. Shekhar, who is pursuing an MS in information science at the Northeastern University, Boston, US. Shekhar worked as a teaching assistant for the robotic science and systems course under Prof. Marslette Vona, helping to build around ten prototype robots with local and global mapping capabilities.

iRobot also has IKA Seaglider UUV having a depth range of 50-1000 metres. It can measure temperature, salinity and other parameters, and send this data back to receiving units using global satellite telemetry. This tool is now being used by the US Navy, government agencies and other research organisations.

"The advantages and disadvantages of both manned and unmanned



Boston Dynamics' military robot dog